

Representation Limitation Module (RLM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Reality Representation Standard (RRS)

Operational Layer: Simulation Reality Representation Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Parent Standard Module

Governed Space: Representation Limitation

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AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Representation Limitation

Canonical Definition

Representation Limitation Module (RLM) defines the governance conditions under which known, discovered, unavoidable, accepted, conditional, and unresolved limitations in a simulation's representation of reality are identified, classified, assessed, bounded, documented, propagated, monitored, and preserved throughout the simulation lifecycle.

RLM establishes the final governance layer of Reality Representation by determining where the internal simulated representation stops being sufficiently representative of the reality it claims to model, what consequences follow from that limitation, and which simulation conclusions, scenarios, conditions, or uses must therefore remain restricted.

RLM does not require a simulation to represent reality perfectly.

It requires the difference between:

Represented Reality

and:

Relevant Reality

to remain governable.

Operational Role

The preceding Reality Representation modules establish:

RSM — What parts of reality must be represented?

RAM — How may selected reality legitimately be simplified?

RFM — How faithfully must the representation correspond to material reality?

RRM — At what resolution must material phenomena remain distinguishable?

RLM asks:

What material limitations remain after reality selection, abstraction, fidelity, and resolution have been established, and how must those limitations constrain subsequent simulation use and interpretation?

The governed progression is:

Selected Reality → Approved Abstraction → Required Fidelity → Required Resolution → Limitation Identification → Limitation Classification → Materiality Assessment → Limitation Boundary → Propagation → Representation Limitation Status

RLM closes the Reality Representation Standard before the simulation proceeds into subsequent SIMULOS® governance spaces.

Module Operational Space

RLM governs: representation limitation identification, known limitations, discovered limitations, inherent limitations, technical limitations, methodological limitations, abstraction limitations, fidelity limitations, resolution limitations, data-dependent representation limitations, environmental representation limitations, behavioral representation limitations, interaction limitations, temporal limitations, spatial limitations, boundary limitations, omitted-reality limitations, simplification effects, representational uncertainty, limitation materiality, limitation severity, limitation applicability, limitation dependencies, limitation interaction, limitation accumulation, limitation acceptance, conditional limitation acceptance, unresolved limitations, limitation propagation, limitation disclosure, limitation monitoring, limitation change, limitation reassessment, limitation status, and limitation traceability.

Module Function

Every simulation represents reality imperfectly.

This is not automatically a defect.

A simulation necessarily selects, simplifies, discretizes, approximates, aggregates, excludes, and transforms aspects of reality.

The governance problem begins when those differences are: unknown, undocumented, misunderstood, materially underestimated, disconnected from simulation outputs, hidden during interpretation, or ignored when simulation results are used.

RLM exists to prevent: simplifications becoming invisible, known representation weaknesses disappearing from later simulation stages, omitted reality being mistaken for irrelevant reality, high fidelity in one area masking serious limitations elsewhere, limitations being described generically without identifying affected outcomes, accepted limitations being interpreted as resolved limitations, computational constraints silently redefining represented reality, representation limitations being lost when simulations are reused, and simulation outputs being interpreted beyond what the represented reality can legitimately support.

Representation Limitation Principle

SIMULOS® establishes:

A limitation does not invalidate a simulation merely because it exists. A limitation becomes methodologically dangerous when its existence, materiality, boundary, or consequence is not governed.

The objective is therefore not:

No Limitations

but:

Known + Assessed + Bounded + Propagated Limitations Representation Limitation

A Representation Limitation is a known or reasonably identifiable condition under which the internal simulation representation does not fully preserve a materially relevant property, behavior, relationship, state, event, variation, or boundary of Relevant Reality.

A limitation may arise from: deliberate simplification, unavailable knowledge, unavailable data, computational constraints, measurement constraints, modeling choices, environmental complexity, unknown behavior, resolution limits, abstraction, incomplete representation, or inherent inability to reproduce the relevant phenomenon.

Limitation Object

A Limitation Object is the specific representation element affected by a limitation.

It may include: a represented entity, environment, behavior, interaction, dependency, process, state, event, population, physical phenomenon, temporal condition, spatial condition, or representation boundary.

Each material limitation should be linked to the object it affects.

Limitation Origin

The origin of a limitation should remain identifiable where possible.

Possible origins include:

Reality Selection

Reality Abstraction

Representation Fidelity

Representation Resolution

Source Data

Modeling Constraint

Computational Constraint

Measurement Constraint

Environmental Uncertainty

Knowledge Gap

or:

Unknown Origin

This distinction matters because different limitation origins may require different downstream treatment.

Known Limitation

A Known Limitation is identified before or during simulation design.

Known limitations should not disappear merely because they were accepted during design.

Acceptance does not eliminate their relevance.

Discovered Limitation

A Discovered Limitation emerges after the representation has been established.

It may arise during: scenario design, simulation execution, output examination, reality correlation, external review, incident analysis, or later operational use.

A discovered material limitation should be propagated backward and forward through affected simulation governance spaces.

Inherent Limitation

An Inherent Limitation exists where the phenomenon cannot reasonably be represented with sufficient completeness under the available simulation approach.

Examples may include: highly uncertain human behavior, unknown future environmental conditions, unresolved physical processes, incomplete causal knowledge, or phenomena whose relevant characteristics cannot yet be measured.

An inherent limitation may still permit simulation.

Its consequences must remain bounded.

Technical Limitation

A Technical Limitation arises from the capabilities of the simulation implementation or infrastructure.

Examples may include: computational capacity, memory limits, solver constraints, platform limitations, maximum agent population, sensor emulation limits, or execution-time constraints.

Technical limitations must not silently redefine the methodological requirements of the simulation.

Methodological Limitation

A Methodological Limitation arises from the representational approach itself.

It may result from: modeling assumptions, selected abstractions, mathematical formulation, approximation method, behavioral model, aggregation strategy, or representational structure.

Abstraction Limitation

An Abstraction Limitation exists where an approved simplification removes distinctions that remain relevant under particular conditions.

An abstraction may therefore be legitimate generally but limited for a particular scenario.

Fidelity Limitation

A Fidelity Limitation exists where representation correspondence to reality is weaker than required for a particular phenomenon, state, interaction, or interpretation.

RFM identifies fidelity conditions.

RLM governs the resulting limitation where the gap remains.

Resolution Limitation

A Resolution Limitation exists where the representation cannot distinguish material phenomena at the required granularity.

RRM identifies and assesses resolution sufficiency.

RLM carries unresolved resolution consequences into the complete Representation Limitation profile.

Omitted-Reality Limitation

Reality excluded from representation may later prove materially relevant.

An Omitted-Reality Limitation exists where an excluded element can materially affect the simulation inquiry.

This does not automatically mean that the original exclusion was unjustified.

It means the exclusion now constrains interpretation.

Boundary Limitation

A simulation boundary may prevent representation of external effects.

A Boundary Limitation may arise where: external systems influence the simulated system, environmental conditions cross the boundary, human actions occur outside the modeled environment, dependencies extend beyond the simulation, or feedback enters from excluded reality.

Behavioral Limitation

Behavioral models may represent only a bounded set of behaviors.

Where real actors or systems may behave outside that set, the difference should remain visible.

Interaction Limitation

An Interaction Limitation exists where relevant interactions are: simplified, aggregated, delayed, omitted, approximated, or represented conditionally.

Interaction limitations can be material even when individual simulation objects are accurately represented.

Environmental Limitation

An Environmental Limitation exists where environmental complexity exceeds the represented environment.

Examples may include incomplete representation of: weather, terrain, infrastructure, traffic, electromagnetic conditions, resource availability, human activity, or operational disturbances.

Temporal Limitation

A Temporal Limitation exists where the representation cannot sufficiently preserve relevant time-dependent behavior.

This may concern: duration, sequence, timing, latency, degradation, seasonality, transition speed, or long-term evolution.

Spatial Limitation

A Spatial Limitation exists where the representation does not sufficiently preserve materially relevant spatial differences.

Data-Dependent Representation Limitation

A representation may be structurally capable of representing a phenomenon while available data prevents that capability from being meaningfully populated.

RLM distinguishes:

Representation Capability

from:

Representation Support

A sophisticated representation built upon weak or incomplete data may remain materially limited.

Representational Uncertainty

A limitation and uncertainty are related but not identical.

Limitation asks:

What can the representation not sufficiently preserve?

Uncertainty asks:

How uncertain are we about the represented condition or its correspondence to reality?

A limitation may exist with low uncertainty.

For example, a known excluded subsystem.

Uncertainty may exist without a known structural limitation.

RLM preserves the distinction.

Limitation Identification

Limitation identification should ask:

Where does the simulation representation materially differ from Relevant Reality in a way that could affect the simulation inquiry?

Identification should consider: excluded reality, abstractions, fidelity gaps, resolution gaps, model boundaries, environmental simplifications, interaction

simplifications, behavioral constraints, data constraints, and known unknowns.

Limitation Materiality

Not every representation difference is material.

A limitation becomes material where it can reasonably affect: simulation behavior, scenario behavior, output interpretation, reality correlation, uncertainty, downstream evidence, intended reliance, or conclusions derived from the simulation.

Materiality Is Purpose-Relative

A limitation may be immaterial for one simulation purpose and critical for another.

For example:

A simplified wind model may be acceptable for:

Long-Term Energy Production Estimation

but materially insufficient for:

Emergency Turbine Load Simulation.

Therefore:

Representation Limitation Materiality = Limitation × Simulation Purpose × Scope × Scenario × Intended Interpretation

conceptually, not numerically.

Limitation Severity

Material limitations may be classified according to their effect.

A practical classification may include:

Minor

The limitation does not materially change the principal simulation interpretation.

Moderate

The limitation affects defined outputs, scenarios, or conditions.

Major

The limitation materially restricts important simulation conclusions.

Critical

The limitation prevents legitimate use of the representation for a material

simulation purpose.

Undetermined

The consequence cannot yet be sufficiently established.

Limitation Applicability

A limitation should identify where it applies.

Applicability may be bounded by: scenario, object, subsystem, environment, population, operational mode, time period, geographic area, failure condition, output, or simulation claim.

Global vs Local Limitation

A limitation may be:

Local

affecting only a defined part of the simulation,

or:

Global

affecting the complete representation or principal simulation inquiry.

Local limitations should not automatically invalidate unaffected representation.

Global limitations should not be hidden by unaffected local strengths.

Conditional Limitation

A limitation may activate only under defined conditions.

For example:

Representation sufficient below wind speed

X

but:

Representation limited above wind speed X.

Conditional limitations should remain linked to the conditions that activate them.

Limitation Interaction

Multiple individually acceptable limitations may interact.

For example:

Moderate Fidelity Limitation Moderate Resolution Limitation Limited
Environmental Representation

may together create a major limitation for one simulation conclusion.

RLM therefore evaluates limitations both individually and collectively.

Limitation Accumulation

A simulation should not assume that individually accepted limitations remain acceptable when accumulated.

RLM asks:

Does the combined limitation profile materially weaken the representation beyond the intended simulation boundary?

Weakest Material Representation Principle

SIMULOS® establishes:

A simulation conclusion cannot legitimately assume stronger representation than the materially limiting representation condition upon which that conclusion depends.

Strong representation elsewhere does not erase a critical limitation.

Limitation Dependency

A limitation may affect other representation elements.

For example:

Environmental Limitation

may affect:

Sensor Representation → Behavioral Response → Interaction → Simulation Output

RLM therefore preserves limitation dependencies where material.

Limitation Propagation

Material representation limitations must propagate to downstream simulation governance.

Conceptually:

Representation Limitation

↓

Assumption Governance

↓

Model & Environment

↓

Inputs & Data

↓

Scenario Design

↓

Simulation Execution

↓

Simulation Output

↓

Reality Correlation

↓

Simulation Evidence

↓

Reliance Boundary

The limitation need not affect every downstream stage.

But affected stages must not remain unaware of it.

No Limitation Laundering Rule

SIMULOS® establishes:

A known material representation limitation may not disappear merely because the simulation proceeds to a later lifecycle stage.

Changing terminology does not resolve the limitation.

Producing outputs does not resolve the limitation.

Successful execution does not resolve the limitation.

Limitation Acceptance

A material limitation may sometimes be accepted.

Acceptance requires a governed basis.

The acceptance should establish: the limitation, affected representation, materiality, consequence, applicable scope, conditions, residual uncertainty, and resulting interpretation boundary.

Accepted ≠ Resolved

SIMULOS® establishes:

Accepted Limitation ≠ Resolved Limitation

Acceptance means the simulation may proceed while the limitation remains explicitly bounded.

Conditional Acceptance

A limitation may be accepted only under defined conditions.

For example:

Accepted for normal operation scenarios

but:

Not accepted for emergency failure scenarios.

Unresolved Limitation

An Unresolved Limitation exists where: materiality cannot be established, consequences remain uncertain, mitigation is incomplete, available evidence is insufficient, or the limitation remains incompatible with the intended simulation inquiry.

Unresolved limitations must remain visible.

Limitation Mitigation

Where appropriate, limitations may be reduced through: expanded reality representation, changed abstraction, improved fidelity, increased resolution, better data, improved model structure, broader environment representation, additional scenarios, or narrower simulation claims.

RLM governs the limitation status.

It does not prescribe one universal mitigation technique.

Residual Limitation

After mitigation, a limitation may remain.

The remaining condition becomes the Residual Representation Limitation.

Residual limitations must be assessed independently of the original limitation.

Limitation Closure

A limitation may be closed only where it is:

Resolved

or:

No Longer Applicable

Closure should not be used merely because the limitation was accepted.

Representation Limitation Status

RLM may establish:

Representation Limitations Acceptable

Representation Limitations Conditionally Acceptable

Representation Partially Limited

Representation Materially Limited

Representation Critically Limited

or:

Representation Limitation Status

Undetermined

The status must remain bounded to the Simulation Purpose, Scope, scenarios, and intended interpretation.

Representation Limitation Profile

The Representation Limitation Profile consolidates material limitations affecting the simulated representation.

It should preserve:

Limitation → Origin → Affected Object → Materiality → Severity → Applicability
→ Dependencies → Acceptance Status → Residual Limitation → Downstream
Consequence

This profile becomes a primary governance output of RLM.

Limitation Disclosure

Material limitations should be disclosed wherever simulation results could otherwise be interpreted more broadly than the representation supports.

Disclosure should be: specific, understandable, linked to affected results, and proportionate to materiality.

Generic statements such as:

“The simulation has limitations.”

are insufficient where specific material limitations are known.

Limitation-to-Output Binding

Where a limitation affects only particular outputs, the limitation should remain bound to those outputs.

This prevents one global disclaimer from obscuring which conclusions remain strong and which are constrained.

Limitation-to-Scenario Binding

Where a limitation applies only to certain scenarios, it should remain linked to those scenarios.

Limitation-to-Condition Binding

Conditional limitations should remain linked to their activation conditions.

Limitation Change

Representation limitations may change as the simulation evolves.

A limitation may:

Appear

Increase

Decrease

Change Scope

Become Conditional

Become Material

Become Non-Material

Be Resolved

or:

Become Unknown

Material change should trigger reassessment.

Limitation Reassessment

Reassessment may be required when: Simulation Purpose changes, Scope changes, Relevant Reality changes, abstraction changes, fidelity changes, resolution changes, model structure changes, new data becomes available, new scenarios are introduced, new outputs are used, reality correlation reveals mismatch, or intended reliance expands.

Representation Change Rule

SIMULOS® establishes:

A limitation assessment is valid only for the representation configuration, purpose, scope, conditions, and interpretation against which it was established.

Material changes may invalidate the previous assessment.

Limitation Versioning

Material limitation profiles should remain version-linked to the relevant simulation representation.

A limitation profile for:

Representation Version 1.2

should not automatically be assumed applicable to:

Representation Version 2.0.

Historical Limitation Preservation

Resolved limitations may remain historically relevant.

They may explain: prior outputs, earlier discrepancies, previous simulation decisions, historical reliance, or later model changes.

RLM therefore preserves material limitation history.

Limitation Traceability

A governed limitation chain should support:

Relevant Reality → Representation Decision → Limitation Origin → Limitation Object → Materiality → Severity → Applicability → Acceptance / Resolution → Residual Limitation → Downstream Effect

This allows an independent reviewer to reconstruct:

what was not represented sufficiently, why, and what consequences followed.

Representation Limitation Boundary

RLM governs limitations of simulation reality representation.

It does not own every limitation affecting a simulation.

Examples:

Representation Limitation

→ RLM

Assumption Limitation

→ subsequent SIMULOS® assumption governance

Input Data Limitation

→ SIMULOS® Inputs & Data governance

Execution Limitation

→ Simulation Execution governance

Validation Limitation

→ VALIDOS®

General Integrity Failure

→ INTEGROS®

This separation prevents RLM from becoming a generic limitation standard for the entire architecture.

Relationship to Reality Selection

RSM determines what reality belongs inside the representation.

RLM governs material consequences where selected or excluded reality creates a representational limitation.

Relationship to Reality Abstraction

RAM governs whether simplification is legitimate.

RLM governs material limitations that remain because of that simplification.

Relationship to Representation Fidelity

RFM governs required correspondence with reality.

RLM governs unresolved consequences where fidelity remains bounded.

Relationship to Representation Resolution

RRM governs required granularity.

RLM governs unresolved consequences where resolution remains limited.

Relationship to ORA™

ORA™ governs actual Operational Reality.

RLM does not redefine operational reality.

It governs where the simulation's representation of that reality remains limited.

Relationship to VALIDOS®

RLM does not determine whether a simulation or simulation-derived claim is formally valid.

VALIDOS® governs validation.

RLM provides a governed representation-limitation basis that may become relevant evidence or a validation constraint.

Relationship to INTEGROS®

RLM does not govern general integrity.

INTEGROS® may govern integrity of representation objects, records, limitation records, and related lifecycle artifacts.

Relationship to ADIT®

Where continuous auditability or evidential reconstruction is required, ADIT® may preserve observation and evidence concerning representation decisions and limitation changes.

Minimum Implementation Framework

1. Identify Representation Limitations

Determine where the simulation representation materially differs from Relevant Reality. 2. Identify Limitation Origin and Object

Establish where each limitation originates and which representation object it affects. 3. Assess Materiality

Determine whether the limitation can materially affect the Simulation Purpose, Scope, scenario, output, or intended interpretation. 4. Classify Severity and

Applicability

Establish how significant the limitation is and where it applies. 5. Assess
Limitation Interaction

Determine whether multiple limitations create a greater combined effect. 6.
Establish Acceptance Status

Determine whether each material limitation is:

Accepted

Conditionally Accepted

Unresolved

or:

Unacceptable 7. Establish Residual Limitation

Determine what limitation remains after any mitigation or representational
change. 8. Propagate Material Limitations

Bind limitations to affected downstream simulation stages, scenarios, outputs,
and interpretation. 9. Establish Representation Limitation Status

Determine the resulting limitation status for the governed representation. 10.
Preserve Traceability and Reassessment

Maintain limitation history and reassess after material representational or
simulation-purpose change.

Governance Outputs

RLM may produce: Representation Limitation Register, Limitation Object Record,
Limitation Origin Record, Known Limitation Record, Discovered Limitation
Record, Inherent Limitation Record, Technical Limitation Record,
Methodological Limitation Record, Abstraction Limitation Record, Fidelity
Limitation Record, Resolution Limitation Record, Omitted-Reality Limitation
Record, Boundary Limitation Record, Behavioral Limitation Record, Interaction
Limitation Record, Environmental Limitation Record, Temporal Limitation
Record, Spatial Limitation Record, Data-Dependent Representation Limitation
Record, Representational Uncertainty Record, Limitation Materiality
Assessment, Limitation Severity Classification, Limitation Applicability
Record, Limitation Interaction Assessment, Limitation Accumulation Assessment,
Limitation Dependency Map, Limitation Acceptance Record, Conditional
Limitation Acceptance Record, Unresolved Limitation Record, Limitation
Mitigation Record, Residual Representation Limitation, Limitation Propagation
Map, Limitation Disclosure Record, Limitation-to-Output Map,
Limitation-to-Scenario Map, Limitation Change Record, Limitation Reassessment
Record, Representation Limitation Status, Representation Limitation Profile,
Limitation Traceability Map, and Governed Representation Limitation Basis.

Use Case 1 — Autonomous Driving Simulation

Scenario

An autonomous-driving simulation represents urban traffic with high vehicle and sensor fidelity but models pedestrian behavior using a limited set of deterministic movement patterns. Application

RLM identifies a:

Behavioral Representation Limitation

The limitation is assessed as relatively minor for:

Basic Lane-Keeping Simulation

but major for:

Pedestrian Collision-Risk Simulation.

Result

The simulation may continue to support the first inquiry.

Its use for the second must remain restricted until pedestrian behavioral representation is improved or the limitation is otherwise appropriately bounded. Use Case 2 — Spacecraft Thermal Simulation

Scenario

A spacecraft thermal simulation accurately represents the spacecraft but uses simplified environmental assumptions for an unusual orbital condition. Application

RLM identifies an:

Environmental Representation Limitation

applicable specifically to the unusual orbital scenario. Result

Normal-orbit simulation outputs remain unaffected.

Outputs for the unusual orbital condition receive an explicit limitation boundary and cannot inherit the representational confidence of normal-orbit scenarios. Use Case 3 — Financial System Stress Simulation

Scenario

A systemic financial simulation models institutional relationships but omits several indirect dependency channels because reliable data is unavailable. Application

RLM identifies:

Omitted-Reality + Data-Dependent + Interaction Limitations

Each limitation alone may appear moderate.

Their interaction is assessed as potentially major under extreme stress scenarios. Result

The limitation profile restricts interpretation of systemic contagion outputs and propagates the limitation into affected scenario and output governance.

Architectural Position

Representation Limitation is the fifth and final internal governance space of the Reality Representation Standard (RRS).

The complete RRS progression is:

Reality Selection → Reality Abstraction → Representation Fidelity → Representation Resolution → Representation Limitation

Together, the five modules answer:

What reality must be represented?

↓

How may it be simplified?

↓

How faithfully must it correspond to reality?

↓

At what granularity must material phenomena remain distinguishable?

↓

What material representational limitations remain and how must they constrain the simulation?

RLM therefore closes the Reality Representation governance space by producing a bounded and traceable representation basis for the next SIMULOS® Parent Standard.

Foundational Principle

A simulation does not become trustworthy by eliminating every representational limitation; it becomes governable when every material limitation remains identifiable, bounded, traceable, and inseparable from the conclusions it can affect.

Canonical Closing Statement

Representation Limitation Module completes the Reality Representation Standard by governing the material differences that remain between Relevant Reality and its simulation representation after reality selection, abstraction, fidelity, and resolution have been established. Through limitation identification, origin attribution, materiality assessment, severity classification,

applicability boundaries, interaction analysis, acceptance governance, residual-limitation assessment, downstream propagation, disclosure, reassessment, versioning, and traceability, RLM prevents known representational weaknesses from disappearing as simulations progress toward scenarios, execution, outputs, evidence, validation, and operational reliance. RLM therefore establishes the governed boundary between what the simulation representation can legitimately support and what remains outside its representational capability.

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Canonical Source

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